Deepseek ai

A Comprehensive Research Biography

14 June 2026

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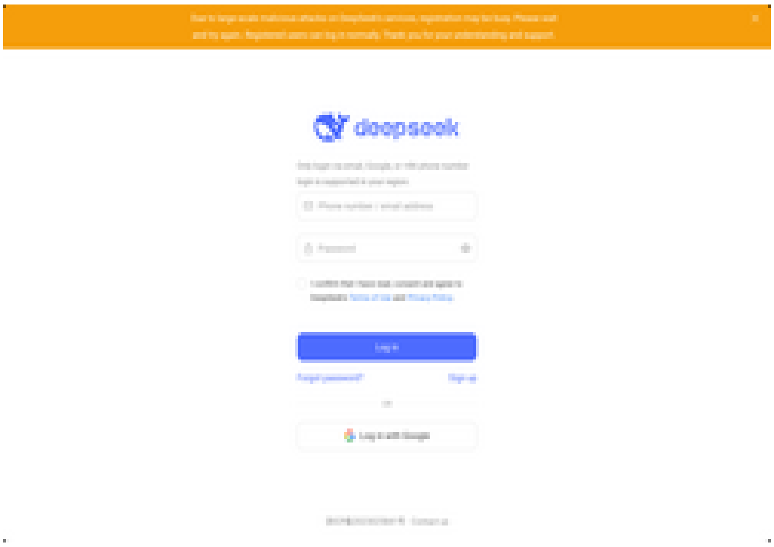
Chapter 1

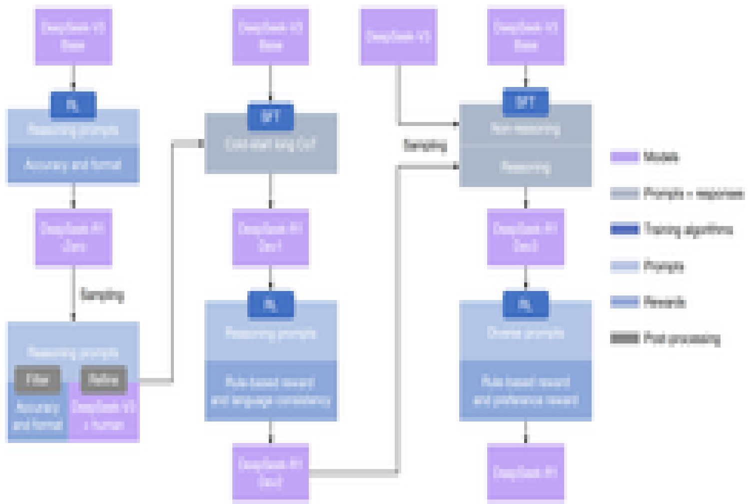
Introduction

Hangzhou DeepSeek Artificial Intelligence Basic Technology Research Co., Ltd., doing business as DeepSeek, is a Chinese artificial intelligence (AI) company that develops large language models (LLMs). Based in Hangzhou, Zhejiang, DeepSeek is owned and funded by High-Flyer, a Chinese hedge fund. DeepSeek was founded in July 2023 by Liang Wenfeng, the co-founder of High-Flyer, who also serves as the CEO for both of the companies. The company launched an eponymous chatbot alongside its DeepSeek-R1 model in January 2025. DeepSeek-R1 provided responses comparable to other contemporary large language models, such as OpenAI's GPT-4 and o1. Its training cost was reported to be significantly lower than other LLMs. The company claims that it trained its V3 model for US$6 million—far less than the US$100 million cost for OpenAI's GPT-4 in 2023—and using approximately one-tenth the computing power consumed by Meta's comparable model, Llama 3.1. DeepSeek's success against larger and more established rivals has been described as "upending AI". DeepSeek's models are described as "open-weight", meaning the exact parameters are openly shared, but the training data is not openly licensed. Since the January 2025 debut of DeepSeek-R1, the company has made its new models available under free and open-source software licenses, primarily the MIT License. The company reportedly recruits AI researchers from top Chinese universities and also hires from outside traditional computer science fields to broaden its models' knowledge and capabilities. DeepSeek significantly reduced training expenses for their R1 model by incorporating techniques such as mixture of experts (MoE) layers. The company also trained its models during ongoing trade restrictions on AI chip exports to China, using weaker AI chips intended for export and employing fewer units overall. Observers say this breakthrough sent "shock waves" through the industry which were described as triggering a "Sputnik moment" for the US in the field of artificial intelligence, particularly due to its open-source, cost-effective, and high-performing AI models. This threatened established AI hardware leaders such as Nvidia; Nvidia's share price dropped sharply, losing US$600 billion in market value, the largest single-company decline in U.S. stock market history.

Chapter 2

Image Gallery





Chapter 3

Career and Contributions

History

Founding and early years (2016–2023)

In February 2016, High-Flyer was co-founded by AI enthusiast Liang Wenfeng, who had been trading since the 2008 financial crisis while attending Zhejiang University. The company began stock trading using a GPU-dependent deep learning model on 21 October 2016; before then, it had used CPU-based linear models. By the end of 2017, most of its trading was driven by AI. Liang established High-Flyer as a hedge fund focused on developing and using AI trading algorithms, and by 2021 the firm was using AI exclusively, often using Nvidia chips. In 2019, the company began constructing its first computing cluster, Fire-Flyer, at a cost of 200 million yuan; it contained 1,100 GPUs interconnected at 200 Gbit/s and was retired after 1.5 years in operation. By 2021, Liang had started buying large quantities of Nvidia GPUs for an AI project, reportedly obtaining 10,000 Nvidia A100 GPUs before the United States restricted chip sales to China. Computing cluster Fire-Flyer 2 began construction in 2021 with a budget of 1 billion yuan. It was reported that in 2022, Fire-Flyer 2's capacity had been used at over 96%, totaling 56.74 million GPU hours. 27% was used to support scientific computing outside the company. During 2022, Fire-Flyer 2 had 5,000 PCIe A100 GPUs in 625 nodes, each containing 8 GPUs. At the time, it exclusively used PCIe instead of the DGX version of A100, since at the time the models it trained could fit within a single 40 GB GPU VRAM and so there was no need for the higher bandwidth of DGX (i.e., it required only data parallelism but not model parallelism). Later, it incorporated NVLinks and NCCL (Nvidia Collective Communications Library) to train larger models that required model parallelism. On 14 April 2023, High-Flyer announced the launch of an artificial general intelligence (AGI) research lab, stating that the new lab would focus on developing AI tools unrelated to the firm's financial business. Two months later, on 17 July 2023, that lab was spun off into an independent company, DeepSeek, with High-Flyer as its principal investor and backer. Venture capital investors were reluctant to provide funding, as they considered it unlikely that the venture would be able to quickly generate an "exit".

Model releases since 2023

DeepSeek released its first model, DeepSeek Coder, on 2 November 2023, followed by the DeepSeek-LLM series on 29 November 2023. In January 2024, it released two DeepSeek-MoE models (Base and Chat), and in April 3 DeepSeek-Math models (Base, Instruct, and RL). DeepSeek-V2 was released in May 2024, followed a month later by the DeepSeek-Coder V2 series. In September 2024, DeepSeek V2.5 was introduced and revised in December. On 20

November 2024, the preview of DeepSeek-R1-Lite became available via chat. In December, DeepSeek-V3-Base and DeepSeek-V3 (chat) were released. On 20 January 2025, DeepSeek launched the DeepSeek chatbot—based on the DeepSeek-R1 model—free for iOS and Android. By 27 January, DeepSeek surpassed ChatGPT as the most downloaded freeware app on the iOS App Store in the United States, triggering an 18% drop in Nvidia's share price. On 24 March 2025, DeepSeek released DeepSeek-V3-0324 under the MIT License. On 28 May 2025, DeepSeek released DeepSeek-R1-0528 under the MIT License. The model has been noted for more tightly following official Chinese Communist Party ideology and censorship in its answers to questions than prior models. On 21 August 2025, DeepSeek released DeepSeek V3.1 under the MIT License. This model features a hybrid architecture with thinking and non-thinking modes. It also surpasses prior models like V3 and R1, by over 40% on certain benchmarks like SWE-bench and Terminal-bench. It was updated to V3.1-Terminus on 22 September 2025. V3.2-Exp was released on 29 September 2025. It uses DeepSeek Sparse Attention, a more efficient attention mechanism based on previous research published in February. DeepSeek-V3.2 was released on 1 December 2025, alongside a DeepSeek-V3.2-Speciale variant that focused on reasoning. In February 2026, Anthropic accused DeepSeek of using thousands of fraudulent accounts to generate millions of conversations with Claude to train its own large language models. In April 2026, investors began speaking with DeepSeek for a $300 million funding round, which would bring DeepSeek to a total valuation of $10 billion. On April 24, 2026, DeepSeek released a preview of its V4 series, including the 1.6-trillion parameter DeepSeek-V4-Pro and the 284-billion parameter DeepSeek-V4-Flash, both featuring a 1-million token context window, under the MIT License. DeepSeek's V4 LLM has been adopted by key semiconductor manufacturers and artificial intelligence chipmakers such as Huawei and Cambricon.

Company operation

DeepSeek is headquartered in Hangzhou, Zhejiang, and is owned and funded by High-Flyer. Its co-founder, Liang Wenfeng, serves as CEO. As of May 2024, Liang personally held an 84% stake in DeepSeek through two shell corporations.

Strategy

DeepSeek has stated that it focuses on research and does not have immediate plans for commercialization. This posture also means it can skirt certain provisions of China's AI regulations aimed at consumer-facing technologies. DeepSeek's hiring approach emphasizes skills over lengthy work experience, resulting in many hires fresh out of university. The company likewise recruits individuals without computer science backgrounds to expand the range of expertise incorporated into the models, for instance in poetry or advanced mathematics. According to The New York Times, dozens of DeepSeek researchers have or have previously had affiliations with People's Liberation Army laboratories and the Seven Sons of National Defence. Due to the impact of United States restrictions on chips, DeepSeek refined its algorithms to maximise computational efficiency and thereby leveraged older hardware and reduced energy consumption. DeepSeek also expanded on the African continent as it offers more affordable and less power-hungry AI solutions. The company has bolstered African language models and generated a number of startups, for example in Nairobi. Along with Huawei's storage and cloud computing services, the impact on the tech scene in sub-saharan Africa is considerable. DeepSeek offers local data sovereignty and more flexibility compared to Western AI platforms.

Training framework

High-Flyer/DeepSeek had operated at least two primary computing clusters: Fire-Flyer ([redacted]) and Fire-Flyer 2 ([redacted]). Fire-Flyer 1 was constructed in 2019 and was retired after 1.5 years of operation. Fire-Flyer 2 is still in operation as of 2025. Fire-Flyer 2 consists of co-designed software and hardware architecture. On the hardware side, Nvidia GPUs use 200 Gbps interconnects. The cluster is divided into two "zones", and the platform supports cross-zone tasks. The network topology was two fat trees, chosen for high bisection bandwidth. On the software side are: 3FS (Fire-Flyer File System): A distributed parallel file system, specifically designed for asynchronous random reads. It uses Direct I/O and RDMA Read. In contrast to standard Buffered I/O, Direct I/O does not cache data. Caching is useless in this case, since each piece of data read is random and is not reused. hfreduce: Library for asynchronous communication, originally designed to replace Nvidia Collective Communication Library (NCCL). It is mainly used for allreduce, especially of gradients during backpropagation. It is asynchronously run on the CPU to avoid blocking kernels on the GPU. It uses two-tree broadcast like NCCL. hfai.nn: Software library of commonly used operators for neural network training, similar to torch.nn in PyTorch. HaiScale Distributed Data Parallel (DDP): Parallel training library that implements various forms of parallelism such as Data Parallelism (DP), Pipeline Parallelism (PP), Tensor Parallelism (TP), Experts Parallelism (EP), Fully Sharded Data Parallel (FSDP) and Zero Redundancy Optimizer (ZeRO). It is similar to PyTorch DDP, which uses NCCL on the backend. HAI Platform: Various applications such as task scheduling, fault handling, and disaster recovery. As of 2022, Fire-Flyer 2 had 5,000 PCIe A100 GPUs in 625 nodes, each containing 8 GPUs. It later incorporated NVLinks and NCCL to train larger models that required model parallelism.

Development and release history

The first DeepSeek models were essentially the same as Llama, which were dense decoder-only transformers. Later models incorporated the multi-head latent attention (MLA), Mixture of Experts (MoE), and KV caching. A decoder-only transformer consists of multiple identical decoder layers. Each of these layers features two main components: an attention layer and a feedforward network (FFN) layer. V2 replaced the standard multi-head attention mechanism (MHA) with multi-head latent attention (MLA). This introduces compressed latent vectors to reduce KV (key–value) cache size, and thus memory usage. A standard MoE Transformer generally use the sparsely-gated MoE layers in the FFN layers. In such an MoE layer, there are several FFN modules in parallel ("routed experts") and a small classifier ("gate") to compute a score for all these modules upon each token. Only the highest-scoring modules are activated. Starting with DeepSeekMoE, DeepSeek adopted a variant that adds "shared experts", which are always activated.

Overview of models and technical specifications

DeepSeek's models are "open weight", which provides less freedom for modification than true open source software.

DeepSeek Coder

DeepSeek Coder is a series of eight models, four pretrained (Base) and four instruction-finetuned (Instruct). All have 16K context lengths. The model was made source-available under the DeepSeek License, which includes "open and responsible downstream usage" restrictions. The training program was: Pretraining: 1.8T tokens (87% source code, 10% code-related English (GitHub markdown and Stack Exchange), and 3% code-unrelated Chinese). Long-context pretraining: 200B tokens. This extends the context length from 4K to 16K. This produced the Base models. Supervised finetuning (SFT): 2B tokens of instruction data. This produced the Instruct models. They were trained on clusters of A100 and H800 Nvidia GPUs, connected by InfiniBand, NVLink, NVSwitch.

DeepSeek-LLM

The DeepSeek-LLM series was released in November 2023. It has 7B and 67B parameters in both Base and Chat forms. DeepSeek's accompanying paper claimed benchmark results higher than Llama 2 and most open-source LLMs at the time. The model code is under the source-available DeepSeek License. The architecture was essentially the same as the Llama series. They used the pre-norm decoder-only Transformer with RMSNorm as the normalization, SwiGLU in the feedforward layers, rotary positional embedding (RoPE), and grouped-query attention (GQA). Both had vocabulary size 102,400 (byte-level BPE) and context length of 4096. They trained on 2 trillion tokens of English and Chinese text obtained by deduplicating the Common Crawl. The Chat versions of the two Base models was released concurrently, obtained by training Base by supervised finetuning (SFT) followed by direct policy optimization (DPO).

V2

In May 2024, DeepSeek released the DeepSeek-V2 series. The series includes 4 models, 2 base models (DeepSeek-V2, DeepSeek-V2 Lite) and 2 chatbots (Chat). The two larger models were trained as follows: Pretrain on a dataset of 8.1T tokens, using 12% more Chinese tokens than English ones. Extend context length from 4K to 128K using YaRN. This resulted in DeepSeek-V2. SFT with 1.2M instances for helpfulness and 0.3M for safety. This resulted in Chat SFT, which was not released. RL using GRPO in two stages. The first stage was trained to solve math and coding problems. This stage used 1 reward model, trained on compiler feedback (for coding) and ground-truth labels (for math). The second stage was trained to be helpful, safe, and follow rules. This stage used 3 reward models. The helpfulness and safety reward models were trained on human preference data. The rule-based reward model was manually programmed. All trained reward models were initialized from Chat (SFT). This resulted in the released version of Chat. They opted for 2-staged RL, because they found that RL on reasoning data had "unique characteristics" different from RL on general data. For example, RL on reasoning could improve over more training steps. The two V2-Lite models were smaller, and trained similarly. DeepSeek-V2 Lite-Chat underwent only SFT, not RL. They trained the Lite version to help "further research and development on MLA and DeepSeekMoE". Architecturally, the V2 models were significantly different from the DeepSeek LLM series. They changed the standard attention mechanism by a low-rank approximation called multi-head latent attention (MLA), and used the previously published mixture of experts (MoE) variant. The Financial Times reported that it was cheaper than its peers with a price of 2 RMB for every million output tokens. The University of Waterloo Tiger Lab's leaderboard ranked DeepSeek-V2 seventh on its LLM ranking. The DeepSeek-Coder V2 series included V2-Base, V2-Lite-Base, V2-Instruct, and V20-Lite-Instruct.. Training: Base models were initialized from corresponding intermediate checkpoints after pretraining on 4.2T tokens (not the version at the end of pretraining), then pretrained further for 6T tokens, then context-extended to 128K context length. DeepSeek-Coder and DeepSeek-Math were used to generate 20K code-related and 30K math-related instruction data, then combined with an instruction dataset of 300M tokens. This was used for SFT. RL with GRPO. The reward for math problems was computed by comparing with the ground-truth label. The reward for code problems was generated by a reward model trained to predict whether a program would pass the unit tests. DeepSeek-V2.5 was made by combining DeepSeek-V2-Chat and DeepSeek-Coder-V2-Instruct.

V3

DeepSeek-V3-Base and DeepSeek-V3 (a chat model) use essentially the same architecture as V2 with the addition of multi-token prediction, which (optionally) decodes extra tokens faster but less accurately. Training process: Pretraining on 14.8T tokens of a multilingual corpus, mostly English and Chinese. It contained a higher ratio of math and programming than the pretraining dataset of V2. Extend context length twice, from 4K to 32K and then to 128K, using YaRN. This produced DeepSeek-V3-Base. SFT for 2 epochs on 1.5M samples of reasoning (math, programming, logic) and non-reasoning (creative writing, roleplay, simple question answering) data. Reasoning data was generated by "expert models". Non-reasoning data was generated by DeepSeek-V2.5 and checked by humans. The "expert models" were trained by starting with an unspecified base model, then SFT on both <problem, original response> data, and synthetic <system prompt, prompt, problem, R1 response> data generated by an internal DeepSeek-R1-Lite model. The system prompt asked R1 to reflect and verify during thinking. Then the expert models were RL using an undisclosed reward function. Each expert model was trained to generate just synthetic reasoning data in one specific domain (math, programming, logic). Expert models were used instead of R1 itself, since the output from R1 itself suffered "overthinking, poor formatting, and excessive length". Model-based reward models were made by starting with a SFT checkpoint of V3, then finetuning on human preference data containing both final reward and chain-of-thought leading to the final reward. The reward model produced reward signals for both questions with objective but free-form answers, and questions without objective answers (such as creative writing). An SFT checkpoint of V3 was trained by GRPO using both reward models and rule-based reward. The rule-based reward was computed for math problems with a final answer (put in a box), and for programming problems by unit tests. This produced DeepSeek-V3. DeepSeek released its DeepSeek-V3-0324 model, which used the same architecture as V3, on 24 March 2025 under the MIT License. The DeepSeek team performed extensive low-level engineering to improve efficiency. They used mixed-precision arithmetic. Much of the forward pass was performed in 8-bit floating point numbers (5E2M: 5-bit exponent and 2-bit mantissa) rather than the standard 32-bit, requiring special GEMM routines to accumulate accurately. They used a custom 12-bit float (E5M6) only for the inputs to the linear layers after the attention modules. Optimizer states were in 16-bit (BF16). They minimized communication latency by extensively overlapping computation and communication, such as dedicating 20 streaming multiprocessors out of 132 per H800 for only inter-GPU communication. They lowered communication by rearranging (every 10 minutes) the exact machine each expert was on so as to avoid querying certain machines more often than others, adding auxiliary load-balancing losses to the training loss function, and other load-balancing techniques. After training, it was deployed on clusters of H800 GPUs. The 8 H800 GPUs within a cluster were connected by NVLink, and the clusters were connected by InfiniBand. The cost has been discussed and called misleading, because it covers only parts of the true cost. Benchmark tests show that V3 outperformed Llama 3.1 and Qwen 2.5 while matching GPT-4o and Claude 3.5 Sonnet.

R1

In January 2025, DeepSeek released the DeepSeek-R1 model under the MIT License. DeepSeek-R1-Lite-Preview was trained for logical inference, mathematical reasoning, and real-time problem-solving. DeepSeek claimed that it exceeded performance of OpenAI o1 on benchmarks such as American Invitational Mathematics Examination (AIME) and MATH. However, The Wall Street Journal reported that on 15 problems from the 2024 edition of AIME, the o1 model reached a solution faster. DeepSeek-R1 and DeepSeek-R1-Zero were initialized from DeepSeek-V3-Base and share its architecture. DeepSeek-R1-Distill models were instead initialized from other pretrained open-weight models, including LLaMA and Qwen, then fine-tuned on synthetic data generated by R1. DeepSeek-R1-Zero was trained exclusively using GRPO RL without SFT. Unlike previous versions, it used no model-based reward. All reward functions were rule-based, "mainly" of two types (other types were not specified): accuracy rewards and format rewards. Accuracy reward was checking whether a boxed answer is correct (for math) or whether a code passes tests (for programming). Format reward was checking whether the model puts its thinking trace within a <think>...</think> tag. R1-Zero has issues with readability and mixing languages. R1 was trained to address these issues and further improve reasoning: SFT DeepSeek-V3-Base on "thousands" of "cold-start" data all with the standard format of |special_token|<reasoning_process>|special_token|<summary>, designed to improve model output readability. Apply the same GRPO RL process as R1-Zero, adding a "language consistency reward" to encourage it to respond monolingually. This produced an un released internal model. Synthesize 600K reasoning data from the internal model, with rejection sampling (i.e. if the generated reasoning had a wrong final answer, then it is removed). Synthesize 200K non-reasoning data (writing, factual QA, self-cognition, translation) using DeepSeek-V3. SFT DeepSeek-V3-Base on the 800K synthetic data for 2 epochs. Apply the same GRPO RL process as R1-Zero with rule-based reward (for reasoning tasks), but also model-based reward (for non-reasoning tasks, helpfulness, and harmlessness). This produced DeepSeek-R1. Distilled models were trained by SFT on 800K data synthesized from DeepSeek-R1, in a similar way as step 3. They were not trained with RL. There were reports that R2, the intended successor to R1, was originally planned for release in early May 2025. However, on 28 May 2025, R1 was instead updated to version R1-0528. As of early July, R2 was not yet released, as Liang Wenfeng was not yet satisfied with its performance. Most Chinese cloud providers of R1 used Nvidia H20. As of August, R2 was not yet released. Sources cite slow data labelling and chip problems. Specifically, DeepSeek was encouraged by authorities to adopt Huawei's Ascend chips for training, but it had stability issues, slower inter-chip connectivity and inferior software. Consequently, it has opted to use Nvidia chips for training and Huawei chips for inference. It is also reported that the Cyberspace Administration of China requested several large corporations to stop buying Nvidia H20 and buy from domestic suppliers instead. With the release of R1 in January 2025, the DeepSeek team published a preprint on arXiv. Later, an updated version was published in Nature in September 2025.

V4

In April 2026, DeepSeek released a preview of their new V4 model series. The DeepSeek V4 model improved on their previous V3/R1 architecture in the following ways: The model uses their Manifold-constrained Hyper Connections (mHC) architecture, which is claimed to "enhance conventional residual connections". They introduced Constrained Sparse Attention (CSA) and Heavily Compressed Attention (HCA), modifications to the Attention mechanism used within Transformer models based on their prior DeepSeek Sparse Attention architecture introduced in V3.2. The Muon optimiser was used for most layers for "faster convergence and improved training stability". They released two sizes of model, V4-Flash and V4-Pro. Each are able to be operated in a non-reasoning mode, a reasoning mode, and a "Max" extended reasoning mode.

Significance

DeepSeek's success against larger and more established rivals was a surprise to both the industry and to markets, and has been compared by investors and pundits to the "Sputnik moment". The DeepSeek-R1 model provides responses comparable to other contemporary large language models, such as OpenAI's GPT-4o and o1. Its training cost is reported to be significantly lower than other LLMs. The company claims that it trained V3, a predecessor of R1, for US$6 million compared to US$100 million for OpenAI's GPT-4 in 2023, and approximately one tenth of the computing power used for Meta's comparable model, LLaMA 3.1. After the January 2025 release of the R1 model, which offered significantly lower costs than competing models, some investors anticipated a price war in the American AI industry. It was dubbed the "Pinduoduo of AI", and other Chinese tech giants such as ByteDance, Tencent, Baidu, and Alibaba cut the price of their AI models. Despite its low price, it was profitable compared to its money-losing rivals.

See also

Artificial intelligence industry in China List of large language models Lists of open-source artificial intelligence software Reasoning model Six AI tigers Six Little Dragons Zhejiang University

External links

Official website DeepSeek on GitHub DeepSeek on Hugging Face Official API documentation Anthology of DeepSeek papers Research blog of High-Flyer

Chapter 4

Recent Developments and Current Work

DeepSeek hiring points to AI infrastructure ambitions beyond rented compute - digitimes

digitimes | 2026-06-13

DeepSeek's hiring activity is drawing almost as much attention as its model releases, with roles for internet data center (IDC) design and planning engineers, senior data center operations engineers and senior delivery managers appearing on major Chinese...

Why Decade-Old Residual Connections Still Power All of AI (And Why That’s a Problem) - Towards Data Science

Towards Data Science | 2026-06-12

For nearly a decade, this part of neural networks barely changed. DeepSeek is trying to reinvent it. Over the past decade, deep learning as a field has grown quite significantly, whether it be the compute capacity of hardware or the ingenuity behind architectures that utilize that hardware. But if you think about it for more than a second, the underlying architecture has remained consistent in a few key areas. We’ve seen a massive shift from convolutional networks to the new Transformer architectures that power today’s large language models, but the way these networks route information from one layer to another hasn’t changed all that much. Recently, researchers at DeepSeek-AI released a paper titled “mHC: Manifold-Constrained Hyper-Connections,” (Xie et al., 2025b) 1 which proposes an entirely new redesign of this routing system. To really appreciate the solution they came up with, let’s look at how signal propagation has evolved over the past few generations of models, and why the current methods are hitting a wall. Firstly, to understand the specific problem that the authors are trying to solve, we need to talk about where it all started–The standard Residual Connection (He et al., 2015) 2 . Introduced back in 2015 with ResNets, the residual connection is arguably one of the most important architectural design choices used in every AI model out there. It simply means that the final output of a layer is the sum of its output and the input it originally got. The key component here is that bare x l term in the residual stream, which we call the identity mapping. It’s important because it acts as an uninterrupted pathway for the gradient signal to flow through the entire network from start to finish. This property is exactly what prevents gradients from vanishing or exploding during training and allows us to successfully train models with hundreds of layers while still ensuring each layer learns and updates itself effectively. But as models have grown increasingly massive, we’ve started to hit the limits of this straightforward approach. In a standard transformer model, we can imagine the residual stream as having a fixed width, which we can refer to as dimension C. Every piece of context, memory, and feature representation has to be crammed into this single C-dimensional vector as it moves up the network. Over time, as the model layers make the information more abstract and expressive, the x l term from the residual stream then becomes the information bottleneck. Typically, if you want to increase the representational capacity of the model, you have to increase the size of the computational layers or add more layers. But by doing that, you also massively increase the compute requirements to run the very model. Because of the above-stated limitation, researchers at ByteDance introduced an alternative to the vanilla residual stream, known as Hyper-Connections (Zhu et al., 2024) 3 . If the normal residual streams are just too "thin", HC widens them. Instead of relying on a single stream of width C , the idea is to expand the width of the residual stream by a specific factor, let's say n . So what you now end up with is a wider vector composed of n parallel streams, resulting in a total width of n×C . But since the actual computational layers of the model, like the Attention and MLP blocks, still expect a standard input with C dimensions only, HC introduces a set of learnable weights to convert the vector between the wide and narrow stream: Fundamentally, by doing this, HC successfully increases the network's capacity and makes the residual stream more expressive. The residual mapping matrix now enables the residual stream to not only allow the unperturbed signal to flow, but also the interactions between the channel dimensions. It allows the model to maintain a much richer internal representation across multiple streams, without increasing the compute cost of the main layers. The reality of the situation, however, is that while HC looks great on paper, it introduces a couple of fatal flaws when you try to scale it up to the size of what our current LLMs are: So, the researchers at DeepSeek were left with a very specific problem: how do you keep the expressive, wide streams of the HC paradigm, without destroying the mathematical stability of the network, and without saturating the GPU memory and I/O operations? To solve these two massive issues prevalent in HC, the DeepSeek team proposed a modified framework which they call Manifold-Constrained Hyper-Connections , or mHC . The solution is broken down into two distinct parts. First, they had to fix the underlying math to stop the signal from exploding/vanishing. Second, they had to do some hardcore systems engineering to make sure the fix could actually run efficiently on modern GPUs. Let's break down exactly how they did both of these. The brilliant mathematical insight here was to take that problematic, unconstrained Residual Mapping matrix and mathematically force it to behave in a constrained manner. To do that, they projected the matrix onto a specific mathematical space known as the Birkhoff polytope. In simpler terms, they constrained the matrix so that it becomes a doubly stochastic matrix . If you aren't familiar with the term, a doubly stochastic matrix is a matrix where all the numbers are non-negative, and every row sums up to exactly 1, and every column also sums up to exactly 1. By forcing the residual matrix into this specific format, authors made sure of a few highly beneficial mathematical properties: To actually turn a regular matrix into a doubly stochastic one during training, the researchers used something called the Sinkhorn-Knopp algorithm (Sinkhorn & Knopp, 1967) 5 . During the forward pass, the algorithm first makes all the numbers in the matrix positive, and then iteratively rescales the rows and columns until they all sum to 1. Solving the math on paper is great, but running all these wide streams and iterative Sinkhorn-Knopp calculations sounds like a nightmare for GPU memory. To get around this, the DeepSeek team implemented some aggressive infrastructure optimizations: Ultimately, the result of all this systems engineering pays off. Despite all the added math and wider streams, mHC only adds a tiny 6.7% time overhead during training compared to a standard baseline model. To see if all the math and system engineering actually paid off, the DeepSeek team put mHC to the test. They trained several language models based on the DeepSeek-V3 architecture (DeepSeek-Ai et al., 2024) 4 , scaling all the way up to a 27-billion parameter model. They compared their new mHC framework directly against a standard residual baseline and the unconstrained, unstable HC paradigm. Let's take a look at how the experiments played out. The main motivation behind mHC was to mitigate the erratic training behaviour that was observed in HC due to the unconstrained mapping matrices. As shown below, the standard HC model's gradient norm (graph b) starts to destabilize with wild swings at around 12k steps, which is exactly the moment where we see the HC and mHC loss plots drift apart (graph a). Because of the smoother and more stable gradient norms with mHC, the model ultimately achieves a lower final training loss when compared to the vanilla HC. A stable model is only useful if it's actually smarter. To prove this, the authors evaluated the 27B variant across multiple downstream benchmarks, including MATH, MMLU, and reasoning tasks like BBH and DROP. As expected, the mHC-enabled model showed consistent performance gains across the board, and especially surpassed the unconstrained HC on a majority of benchmarks. The reasoning benchmarks saw a particularly nice gain in performance, indicating that the wider residual streams are actively contributing to a more expressive model. An important test for any new deep-learning architectural paradigm is if it obeys the pre-established scaling laws or not. Some design choices which work for a 3B parameter model might fail or backfire for a 27B parameter model. To ensure this, the authors plotted the compute scaling curves for 3B, 9B, and 24B parameter models. The below shown graphs clearly demonstrate that the relative loss improvement is maintained across all scales, validating that mHC is a scalable architectural upgrade. As a final test, the authors also tested one of their claims directly: that the signal should not explode arbitrarily when stacked under multiple layers. For the standard unconstrained HC, we saw how the signal can be amplified by a factor of 3,000, which threw the gradients off completely during training. To see if mHC fixed this issue directly or not, DeepSeek tracked the signal propagation dynamics layer-by-layer in the model, and the results were as expected. Due to the doubly-stochastic mapping matrices, the signal gain was capped at around 1.6 throughout the model, proving that the signal

AI brands as bait: How threat actors are using the AI hype in social engineering - Microsoft

Microsoft | 2026-06-08

As threat actors operationalize AI to accelerate attacks, they are also leveraging the wider global interest around AI itself as a social engineering lure.

What DeepSeek’s US$7 billion fundraise means for Hong Kong - South China Morning Post

South China Morning Post | 2026-06-09

Readers discuss an international gateway for China’s AI ecosystem, energy-efficient AI models, a crackdown on distracted driving, and today’s Hong Kong But the most revealing number is the 20 billion yuan founder Liang Wenfeng is said to be investing personally, nearly 40 per cent of the round. That is not just skin in the game. That is a declaration that DeepSeek will remain a research-first, open-source laboratory, not a commercial product factory. The strategic logic departs from Silicon Valley orthodoxy in ways the West has yet to fully grasp. DeepSeek’s models are open-weight, its developer community spans the globe, and its success is measured in GitHub stars and third-party integrations, not revenue. This is not altruism – it is platform-building by other means. When start-ups in Jakarta and universities in Nairobi build on Chinese open-source stacks, they are building on Chinese architecture. The default becomes the destination.

Hackers are capitalizing on AI hype to ramp up social engineering attacks – and they're using big brands like Anthropic, OpenAI, and DeepSeek as ‘bait’ to lure victims - IT Pro

IT Pro | 2026-06-11

Cyber criminals are exploiting AI hype to impersonate the branding of AI platforms such as ChatGPT, Microsoft Copilot, DeepSeek, and Anthropic’s Claude, according to new research. Microsoft Threat Intelligence said it's observed an uptick in phishing , malvertising, and search engine optimization (SEO)-driven attacks that ultimately lead to credential theft, financial fraud, or malware infection. Campaigns focus on highly anticipated launches or emerging trends, using tried-and-tested tactics such as urgency-driven messaging, abuse of trusted services, and multi-stage redirection chains that require user interaction to evade detection. "While traditional lures like invoices, payment notifications, or delivery alerts remain effective and continue to be widely used, AI-themed lures reflect a shift in social engineering that is likely to persist as a long-term tactic used by threat actors, from cyber criminal groups to nation states," the company warned. In one example, Microsoft said it had observed a ChatGPT-themed phishing attack delivering malicious URLs which led to phishing pages that collected credit card and personal information such as names and addresses. The emails used the sender display name ChatGPT and the subject line: “To ensure your ChatGPT Plus continues to work – please update your payment method”. This phishing activity, which consisted of 4,500 emails sent to targets in South Africa, was part of a broader campaign using similar themes and infrastructure that delivered as many as 100,000 emails on a single day to targets in Switzerland, Austria, and South Africa. Microsoft noted the campaign affected a broad range of industries, including higher education and professional services. In another example, security experts spotted a phishing campaign impersonating Anthropic-branded services to target users with account-related lures tied to the Claude AI platform. The campaign sent phishing emails to targets across more than 2,000 organizations, mainly in the US, UK, and India. "The campaign used enforcement-themed messaging claiming that the recipient’s account was in violation of acceptable use policies and required immediate action," the company noted. "The emails impersonated Anthropic’s popular AI service Claude using the display names Anthropic Teams and Anthropic PBC, masquerading as legitimate account-related communications. Subject lines followed a consistent structure of 'Claude Appeal Request' combined with date elements." Other examples included malvertising campaigns that use AI-themed terms such as 'Awesome AI Windows Plugin' and 'Flux Pro AI' in social engineering lures, and fake DeepSeek V4 installers on GitHub that delivered Vidar Stealer. "Within hours of DeepSeek previewing their latest version, V4, attackers created a fake GitHub organization and repository. They copied real branding and benchmark data, added AI and SEO-search-friendly content, and pushed malicious archives that looked like installers," explained John Bruggeman, vCISO at CBTS. "What the attacker did was not particularly exotic, but it was well timed and convincingly packaged. A user searching for the newest model could very easily end up in the wrong place, especially because the malicious repository showed up in GitHub, Google, Bing, or AI-assisted search results. The search results added legitimacy to the malware ." To counter these rising threats, Microsoft advised customers to configure automatic attack disruption in Microsoft Defender XDR, enforce multi-factor authentication (MFA) on all accounts, use the Microsoft Authenticator app for passkeys and MFA, and scope conditional access policies to strengthen privileged accounts with phishing-resistant MFA . "The companies that have a handle on AI governance (policies and procedures) well will be the ones that make safe AI use easy, risky AI use visible, and malicious activity hard to ignore. That means publishing a clear list of approved tools, blocking obvious lookalike domains and very recently registered domains can help stop this kind of threat," said Bruggeman. "Monitoring suspicious downloads and sign-ins, and training employees on the AI-themed lures should also be done right now - don't think that generic phishing examples from five years ago are going to cut it today." Follow ITPro on Google News and add us as a preferred source to keep tabs on all our latest news, analysis, views, and reviews. Emma Woollacott is a freelance journalist writing for publications including the BBC, Private Eye, Forbes, Raconteur and specialist technology titles. 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